# Day 04 | Tuesday

04

November 3, 2026

## Brackets, squares and square roots

12:45 PM–2:00 PM • Math • 75 min

## Calculation checking workshop

2:15 PM–3:30 PM • Math • 75 min

**Week 2 • 2 scheduled blocks • 150 teaching minutes**

Estimate. Show the method. Label units. Check the result.

# Today’s exact schedule

## 12:45 PM–2:00 PM • 75 min

### Brackets, squares and square roots

| | | |
|---|---|---|
| 12:45 PM–12:50 PM | **Start** | 5 |
| 12:50 PM–12:55 PM | **Recall** | 5 |
| 12:55 PM–1:10 PM | **Teach** | 15 |
| 1:10 PM–1:25 PM | **Together** | 15 |
| 1:25 PM–1:45 PM | **Practise** | 20 |
| 1:45 PM–1:55 PM | **Check** | 10 |
| 1:55 PM–2:00 PM | **Wrap up** | 5 |

Math • Math Week 2, Lesson 3

## 2:15 PM–3:30 PM • 75 min

### Calculation checking workshop

| | | |
|---|---|---|
| 2:15 PM–2:20 PM | **Start** | 5 |
| 2:20 PM–2:25 PM | **Recall** | 5 |
| 2:25 PM–2:40 PM | **Teach** | 15 |
| 2:40 PM–2:55 PM | **Together** | 15 |
| 2:55 PM–3:15 PM | **Practise** | 20 |
| 3:15 PM–3:25 PM | **Check** | 10 |
| 3:25 PM–3:30 PM | **Wrap up** | 5 |

Math • Math Week 2, Lesson 4

**Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time**

# Brackets, squares and square roots

**Today’s goal: Explain what an exponent or root acts on and avoid applying it to the wrong part of an expression.**

Name one use for today’s skill.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

Exercise Book 3A Q4–10 (p.19); 2A Q2–3 (p.13)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Recall and reconnect

Discuss whether $(6 - 4)^2$ means the same as $6^2 - 4^2$. Explain before calculating.

**Try first without displayed answers.**

Resources: Exercise Book 3A Q4–10 (p.19); 2A Q2–3 (p.13)

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Identify the base of an exponent

**1** In $4^2$, only the 4 is squared.

**2** In $(6 - 4)^2$, the complete bracket result is squared.

**3** Calculate inside brackets before applying the exponent.

**4** A changed bracket can change the value even when the same numbers appear.

$4^2$ **base: 4**

$(6 - 4)^2$ **base: the whole bracket (6 − 4)**

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

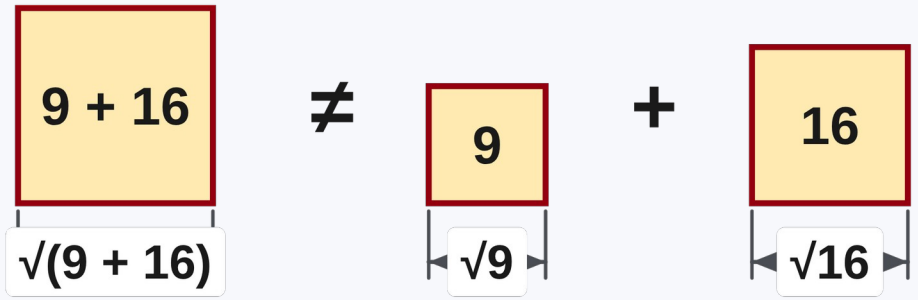
# Identify the whole quantity under the root

1 The root sign acts on the quantity written under it.

2 For √(9 + 16), add inside the grouping before finding the square root.

3 A square root of a sum is not generally the sum of the separate roots.

4 Write the grouping before using your calculator’s model-specific root sequence.

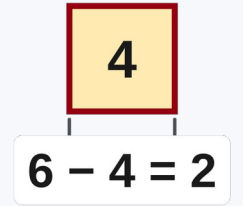



Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

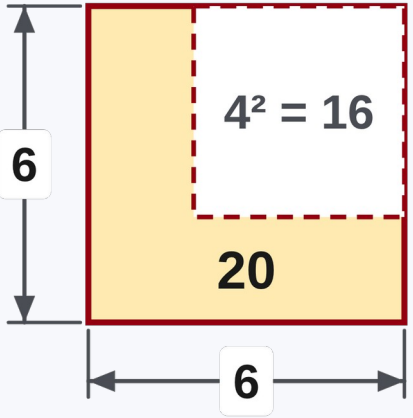
# Mark the base before squaring

$(6 - 4)^2 = 2^2 = 4$
$6^2 - 4^2 = 36 - 16 = 20$
$3 \times 4^2 = 3 \times 16 = 48$

Square root check: √(9 + 16) = √25 = 5.
It is not √9 + √16, which equals 7.




$(6 - 4)^2 = 4$




$6^2 - 4^2 = 20$

Together • Exercise Book 3A Q4–7.
Say the next operation aloud; compare paper and calculator results.
Use Numeracy Hub Common mistakes only for taught examples.

# Textbook | page 47

## Example 3.5 What does the square belong to?

Find $100 - 5^2$, $(100 - 5)^2$, $5 \times 3^2$ and $(5 \times 3)^2$.

**SOLUTION**

1. $100 - 5^2$: the square belongs to the 5. $5^2 = 25$, then $100 - 25 = 75$.
2. $(100 - 5)^2$: brackets first, 95, then $95^2 = 9{,}025$.
3. $5 \times 3^2$: the square belongs to the 3. $3^2 = 9$, then $5 \times 9 = 45$.
4. $(5 \times 3)^2$: brackets first, 15, then $15^2 = 225$.

Textbook p.47 · Example 3.5: identify what the square belongs to.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Textbook | page 48

## Example 3.6 A root over a sum

Find √(9 + 16) and √9 + √16.

**SOLUTION**

1. √(9 + 16): the root covers the sum. 9 + 16 = 25, then √25 = 5.
2. √9 + √16: two separate roots. √9 = 3 and √16 = 4, then 3 + 4 = 7.
3. On the Sharp, key √ and then the sum in brackets. On the TI-30Xa, key the sum in brackets, then √x after the closing bracket.

Textbook p.48 · Example 3.6: one root over a sum versus two separate roots.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Spot the mistake. Explain the correction.

## A learner says $(6 - 4)^2$ and $6^2 - 4^2$ must have the same value. What changes?

Discuss the method before checking the answer.

**The first squares the difference and gives 4. The second subtracts two squares and gives 20.**

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Work together

Work 3A questions 4–7. Identify the next operation aloud and compare paper and calculator results.

**Work in pairs; alternate solver and checker.**

Resources: Exercise Book 3A Q4–10 (p.19); 2A Q2–3 (p.13)

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Guided practice | 3A Q4

$(15 - 7) \times 4$

$(15 - 7) \times 4$ — **brackets first**

$= ? \times 4$ — **× 4 carried forward**

$= ?$

Work with a partner. Take turns explaining the next step.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Compare your method | 3A Q4

$$(15 - 7) \times 4$$

**CHECK TOGETHER**

**32**

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Your scheduled practice

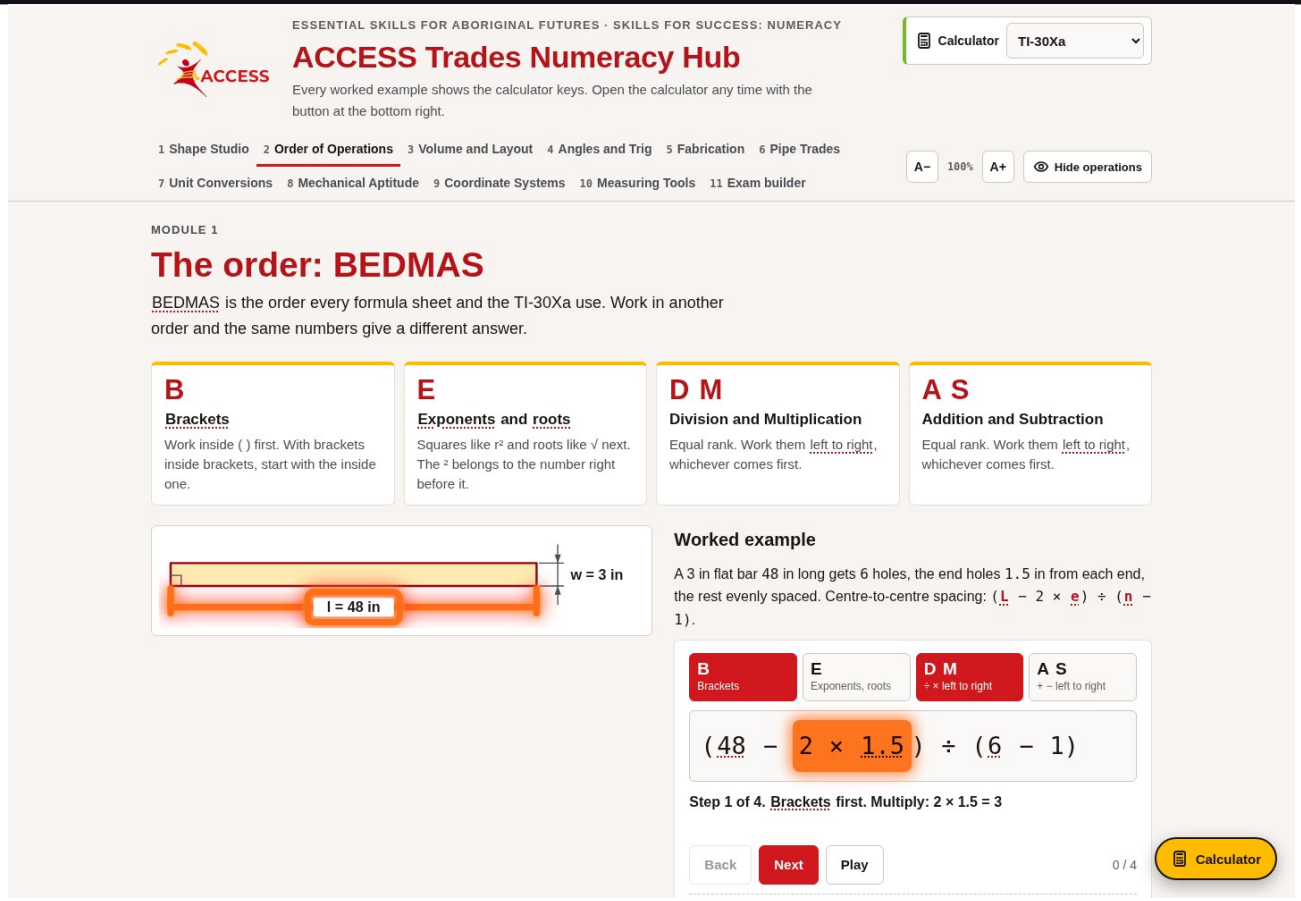
Solve 3A questions 8–10 and 2A questions 2–3. Make a deliberate bracket error, explain it, then correct it.

**Show working and units; estimate, then check.**

Resources: Exercise Book 3A Q4–10 (p.19); 2A Q2–3 (p.13)

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Open Numeracy Hub




Navigation example. Use the values and units in the assigned task.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Practise in Numeracy Hub

1. Supported: Open Numeracy Hub → Order of Operations → Common mistakes; use taught examples only.

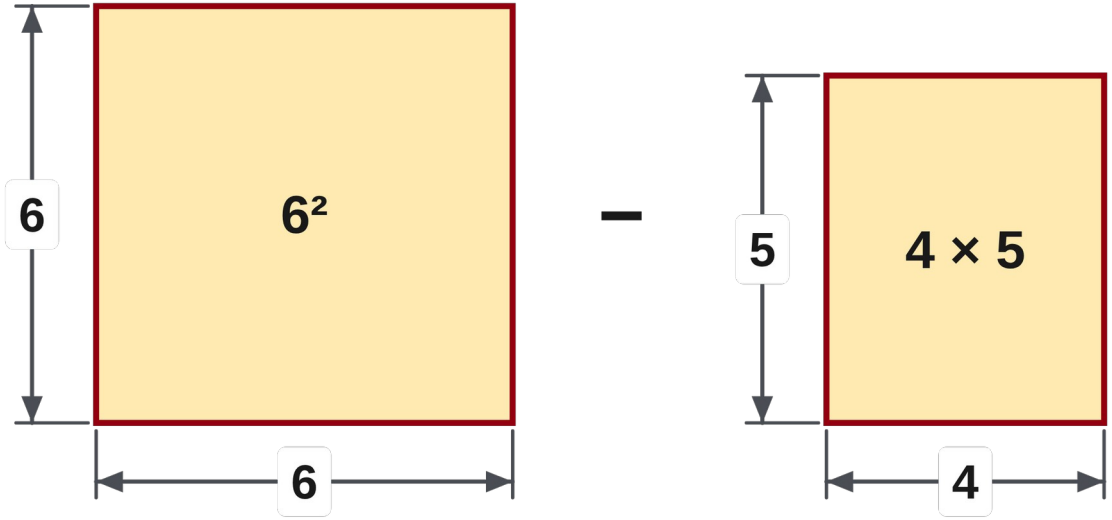
2. Independent practice: Write a deliberate bracket error beside a corrected version; explain the scope.

3. Independent checking: Number the operations and compare the correct working with the tool.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Practise from the Exercise Book | 3A Q8

$6^2 - 4 \times 5$

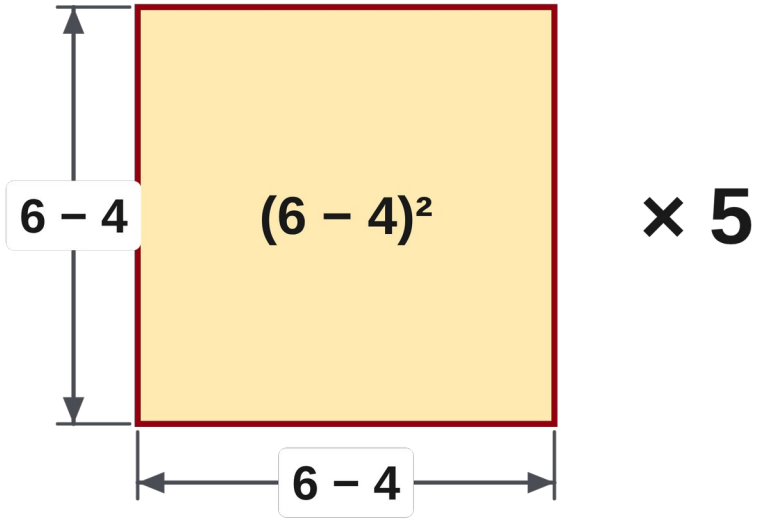



Write the operation, working and units before checking.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Practise from the Exercise Book | 3A Q9

$(6 - 4)^2 \times 5$




Write the operation, working and units before checking.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Check independently

Complete 3A questions 8–9 with each operation numbered.

**Answer independently and show the method.**

Resources: Exercise Book 3A Q4–10 (p.19); 2A Q2–3 (p.13)

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Independent check | 3A Q8

$6^2 - 4 \times 5$

Answer independently. Keep worked solutions closed.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Independent check | 3A Q9

$$(6 - 4)^2 \times 5$$

Answer independently. Keep worked solutions closed.

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Wrap up and choose the next step

Share one learning point and one skill to practise next.

**Record a takeaway and next action.**

Resources: Exercise Book 3A Q4–10 (p.19); 2A Q2–3 (p.13)

Textbook Chapter 3 pp.45–51; Chapter 2 pp.31–33

# Break

## 2:00 PM–2:15 PM

Return ready for: Calculation checking workshop

**This break is excluded from teaching minutes.**

# Calculation checking workshop

**Today’s goal: Use estimation, inverse operations and a second method to check calculations.**

Name one use for today’s skill.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

Exercise Book 1A Q20 (p.7); 3B Q2–3 (p.21); 3C Q2–3 (p.23)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Recall and reconnect

Take a moment to name one common keying error and one way to detect it.

**Try first without displayed answers.**

Resources: Exercise Book 1A Q20 (p.7); 3B Q2–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Choose a check for the error you suspect

1. An estimate checks whether the answer is a sensible size.
2. An inverse operation checks the numerical relationship.
3. A second method checks whether the original setup counted the right quantities.
4. Repeating the same key sequence can repeat the same error.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

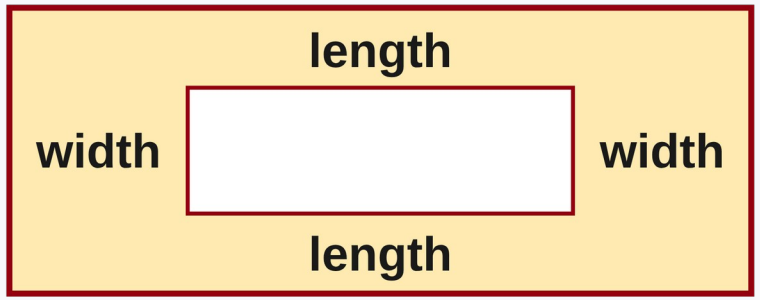
# Put a missing dimension back into the total

**1** Write the original perimeter relationship before solving for a missing side.

**2** Substitute the found dimension into that original relationship.

**3** Check that all four sides recreate the stated perimeter.

**4** Keep an estimate and a separate check line beside the solution.




**all four sides = stated perimeter**

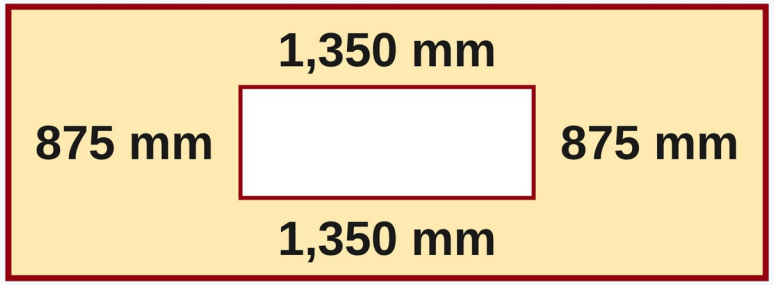
Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Use a genuinely different method

Exercise Book 3B Q2: rectangle sides 1,350 mm and 875 mm.

**1,350 + 875 + 1,350 + 875 = 4,450 mm**
**2 × (1,350 + 875) = 4,450 mm**

Both count the same four sides. An estimate of 2 × (1,400 + 900) = 4,600 mm supports the scale.




Together • Recheck 3B Q2 both ways.
Then substitute a missing dimension back into the original perimeter. Ask whether the total is restored.

W2 L4 • Repeating an incorrect key sequence is not an independent check

# Textbook | page 39

## 2.11 Checking a result

A calculator does exactly what you key, including your mistakes. Before you cut or order, check the answer in at least one of these ways.

1. Estimate first. Round the numbers and work it out in your head, as in Chapter 1. The calculator answer should be close.
2. Work it backwards. If $\sqrt{144} = 12$, then $12^2$ should give 144. If 6,000 − 1,625 = 4,375, then 4,375 + 1,625 should give 6,000.
3. Read the Sharp top line. It shows the calculation you keyed, so you can spot a wrong number before you press =.
4. Key it again. If the second answer differs from the first, key it a third time slowly.

Textbook p.39 · Checking a result: estimate, reverse and inspect the entry.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Textbook | page 39

### WRITE THE ANSWER WITH ITS UNIT

The display shows a number, not a size. Write the unit next to every answer you copy down: 4,375 mm, not 4375. A number without a unit is a guess on the shop floor.

Textbook p.39 · Record the unit with the checked answer.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Spot the mistake. Explain the correction.

## Two attempts with identical calculator keys give the same number. Is that a different checking method?

Discuss the method before checking the answer.

**No. Check with an inverse operation or a different setup, such as adding sides instead of grouping pairs.**

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Work together

Recheck 3B question 2 in two ways: add the four sides and use the grouped formula.

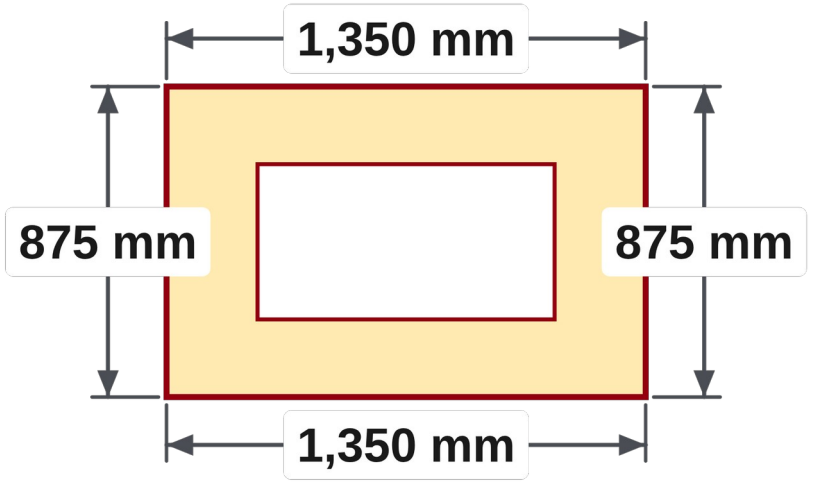
**Work in pairs; alternate solver and checker.**

Resources: Exercise Book 1A Q20 (p.7); 3B Q2–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Guided practice | 3B Q2

A rectangular guard frame is 1,350 mm by 875 mm. The distance around it is 2 × (1,350 + 875). Find it.




**Distance around:** ?

Work with a partner. Take turns explaining the next step.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Compare your method | 3B Q2

A rectangular guard frame is 1,350 mm by 875 mm. The distance around it is 2 × (1,350 + 875). Find it.

**CHECK TOGETHER**

## 4,450 mm

Brackets first: 2,225. Then 2 × 2,225.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Your scheduled practice

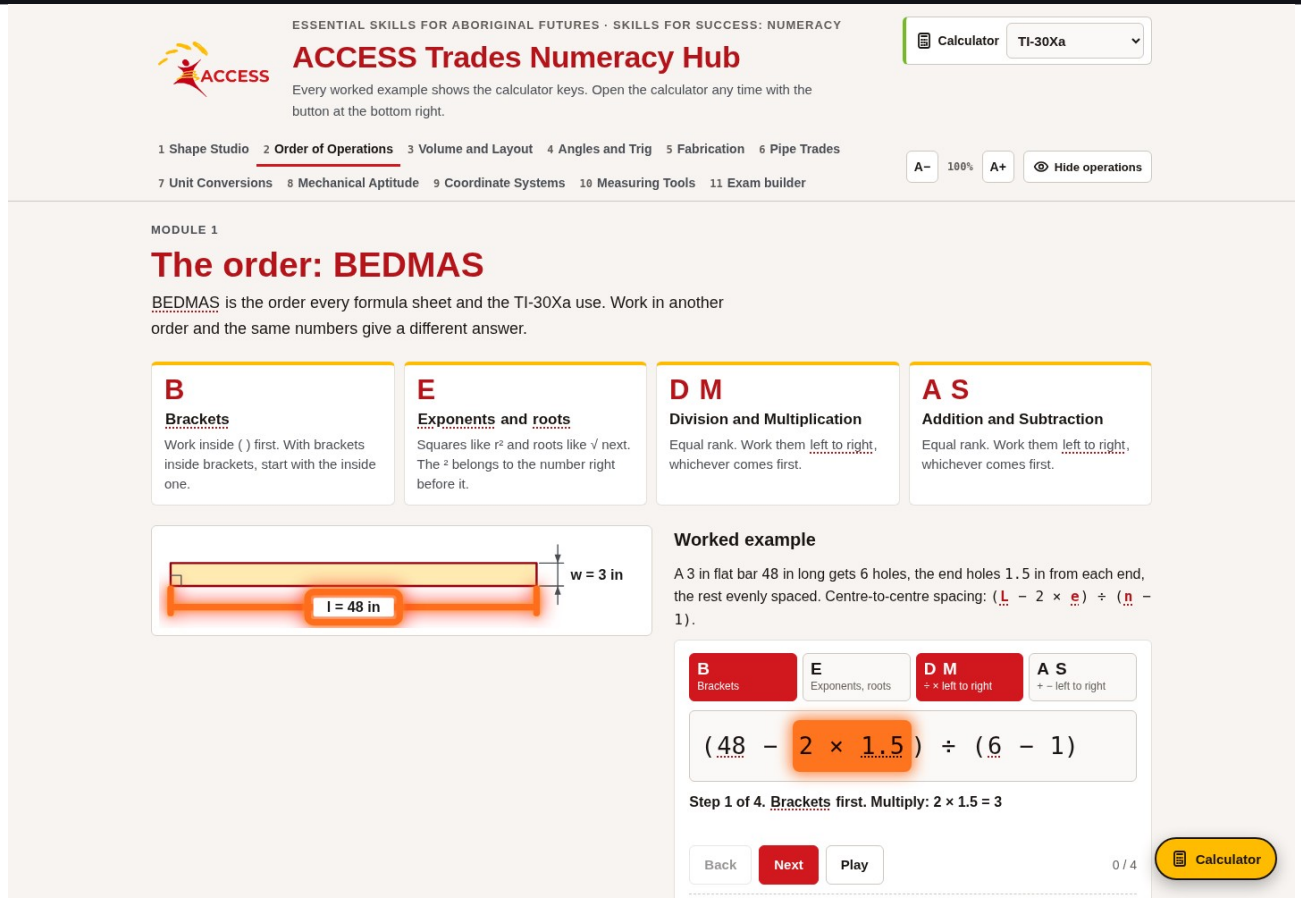
Complete 3B question 3 and 3C questions 2–3. Add an estimate and a check line to every solution. Finish with two new Numeracy Hub questions.

**Show working and units; estimate, then check.**

Resources: Exercise Book 1A Q20 (p.7); 3B Q2–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Open Numeracy Hub




Navigation example. Use the values and units in the assigned task.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Practise in Numeracy Hub

1. Supported: For 3B Q2, compare adding four sides with the grouped perimeter formula.

2. Independent practice: Solve two Easy Numeracy Hub BEDMAS questions from taught topics on paper.

3. Independent checking: Keep each question and add an estimate or second-method check before review.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Practise from the Exercise Book | 3B Q3

Four 27 in pieces and two 15 in pieces are cut from a 144 in bar. Ignore the saw cuts. The offcut is 144 − (4 × 27 + 2 × 15). Find it.

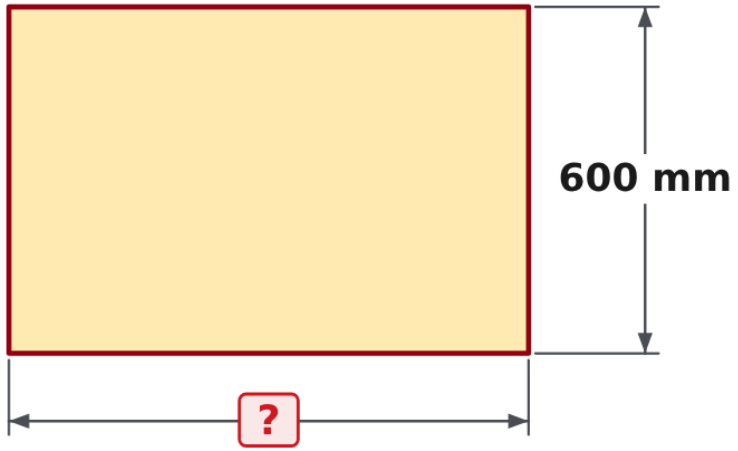



Write the operation, working and units before checking.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Practise from the Exercise Book | 3C Q2

The distance around this frame is 3,000 mm. Find the hidden length with $L = 3{,}000 \div 2 - 600$.




Write the operation, working and units before checking.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Check independently

Explain why 3C question 2 is 900 mm and show that it gives the stated 3,000 mm perimeter.

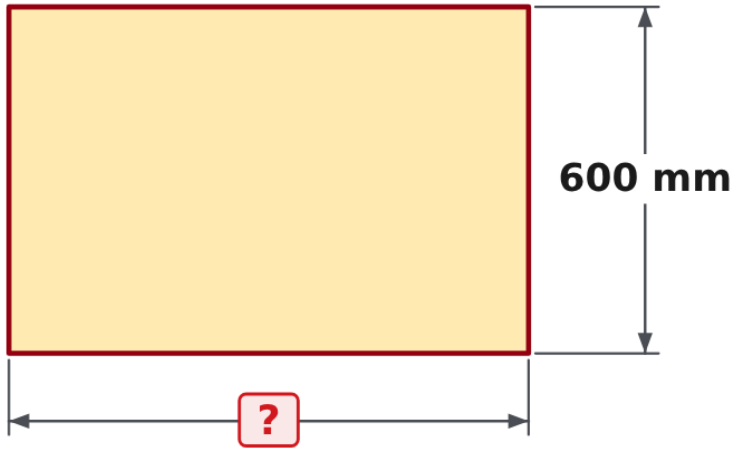
**Answer independently and show the method.**

Resources: Exercise Book 1A Q20 (p.7); 3B Q2–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Independent check | 3C Q2

The distance around this frame is 3,000 mm. Find the hidden length with $L = 3{,}000 \div 2 - 600$.




Answer independently. Keep worked solutions closed.

Textbook Chapter 2 p.39; Chapter 3 pp.44–51

# Wrap up and choose the next step

Share one learning point and one skill to practise next.

**Record a takeaway and next action.**

Resources: Exercise Book 1A Q20 (p.7); 3B Q2–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 2 p.39; Chapter 3 pp.44–51